

Prajwal Navada G P

prajwalnavada74@gmail.com | Bengaluru, Karnataka | LinkedIn | GitHub | Portfolio

SUMMARY

Second-year CSE student (CGPA 9.2) with experience building software systems, real-time applications, and Edge AI solutions across web, IoT, and embedded domains. Developed end-to-end projects involving distributed communication, real-time synchronization, TinyML deployment, predictive analytics, and cloud-connected embedded systems. Proficient in React, Node.js, MySQL, and Socket.io; hands-on with ESP32, TinyML (Edge Impulse), MQTT, and sensor fusion. Building at the intersection of modern software and intelligent hardware — targeting internships in software engineering, IoT, and Edge AI.

EDUCATION

Dr. H.N. National College of Engineering, Bengaluru

B.E. Computer Science & Engineering · Visvesvaraya Technological University

2024 – 2028

CGPA: 9.2

TECHNICAL SKILLS

Programming:	Java, JavaScript, Python, C, C++
Software Engineering:	React, Vite, Node.js, Express.js, REST APIs, JWT Authentication, Socket.io, Spring Boot
Databases & Backend:	MySQL, Database Design, Firebase Cloud Messaging, Nodemailer, Google Drive API
Embedded & Edge AI:	ESP32, Arduino, MQTT, TinyML (Edge Impulse), ThingsBoard, DHT11, MPU6050, Sensor Fusion
Systems & Tools:	Linux (Ubuntu), Git, GitHub, PWA (Workbox), Postman, Agile
Core CS:	DSA, OOP, DBMS, Computer Networks, Operating Systems, Real-Time Systems, Predictive Analytics

PROJECTS

EdgeAI-Predict — TinyML Predictive Maintenance System

GitHub

ESP32 · TinyML (Edge Impulse) · MQTT · ThingsBoard · DHT11 · MPU6050 · Relay

- Engineered a complete **Edge AI monitoring system** on a single ESP32, eliminating dependency on gateway devices while performing real-time anomaly detection and predictive maintenance.
- Designed a **hybrid intelligence pipeline** combining signal processing (Moving Average), Linear Regression forecasting, rule-based decision logic, and TinyML inference for proactive fault detection.
- Implemented secure **MQTT-based telemetry** and remote actuation workflows integrated with a ThingsBoard dashboard for real-time monitoring, historical trends, predicted values, and bidirectional control.
- Optimized memory usage, inference execution, and sensor-processing workflows to ensure reliable operation on resource-constrained ESP32 hardware.

Auditorium Booking System

GitHub

React · Vite · Node.js · Express · MySQL · Firebase FCM · PWA

- Architected a **multi-role booking platform** for 3 colleges supporting approval workflows, scheduling automation, conflict-free allocation, and audit logging for operational transparency.
- Implemented role-based authorization, automated notification pipelines (email + Firebase push), scheduled background cron jobs, and **role-scoped PDF/Excel reporting**.
- Designed **scalable database structures** supporting unlimited users across Admin, Staff, and Principal roles — PWA support for installable, offline-capable experience.

QuizLive — Real-Time Quiz & Assessment Platform

GitHub | Live

React · Node.js · Express · Socket.io · MySQL · JWT

- Built a **real-time multiplayer assessment platform** with synchronized quiz sessions, dynamic leaderboards, streak multipliers, and **sub-100ms score propagation** for concurrent players.
- Developed secure JWT authentication, attempt tracking, and anti-cheat mechanisms using session validation and client activity monitoring (tab-switch/focus-loss detection).
- Designed **10-table normalized relational schema** and backend services supporting concurrent users and real-time event processing.

Smart Study — Academic Resource Portal

GitHub | Live

React · Vite · Node.js · Express · Google Drive API

- Developed a centralized academic content platform for **10+ subjects** with structured access to notes, study materials, and previous-year exam resources via Google Drive API with protected downloads.
- Implemented **automated catalog reconstruction** capabilities to recover resource indexes from Drive metadata after database loss — no manual intervention required.
- Deployed on Render with admin dashboard for live JSON snapshots; zero-downtime re-indexing on data loss events.

CERTIFICATIONS

Programming in Java — NPTEL · IIT Kharagpur | Government-backed certification: Java fundamentals, OOP, and core CS concepts.

ACHIEVEMENTS & LEADERSHIP

- Won **all intra-college quiz competitions** participated in — consistently ranked among top performers in time-constrained problem-solving environments.
- **Led 3+ student project teams** — coordinating planning, implementation, testing, and on-time delivery across software and embedded initiatives.
- Active contributor to technical communities, peer-learning initiatives, and collaborative engineering projects at college.